Communication Networks - Lab 2 Report

Group ID: 13

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Preliminary Questions

1. **Characteristic parameters:** Characteristic parameters of a PCM system are the number of bits (n_{bit}) for the ADC and the error probability (P_e) for the BSC. Other parameters involve the analog signal bandwidth and the sampling frequency.
2. **Final aim:** System parameters are determined based on the goal of assessing and achieving a good reconstruction of the input signal. For example, parameter decisions are driven by the need to meet a specific quantization SNR requirement after DAC reconstruction.
3. **Meaning of e_Q and e_B (e_{BSC}):** In the provided block diagram, e_Q represents the quantizing equivalent noise added to the signal. The term e_B (or e_{BSC}) represents the BSC equivalent noise resulting from bit errors during transmission.
4. **Quality metric:** The primary quality metric used to assess the successful reconstruction of the analog signal is the Signal-to-Noise Ratio (SNR).

1. Introduction

- **Objective:** Primary goal of the lab is to create the software representation of a PCM system.
- **Theoretical Background:** A PCM system can be modeled as a system which comprises three parts: ADC, BSC, DAC. The ADC is responsible for taking an analog signal and converting it into a digital one. The DAC performs the reverse operation. The BSC mathematically models the transmission of these generated bits over a channel, which introduces a certain error probability (P_e) of changing a bit's state.
- **Connection to Class:** This lab directly applies theoretical concepts by calculating the total Signal-to-Noise Ratio (SNR) of the modeled system. We analyze how the overall PCM system SNR acts as the parallel combination, in linear units, of the quantization SNR and the BSC SNR.

2. Methodology

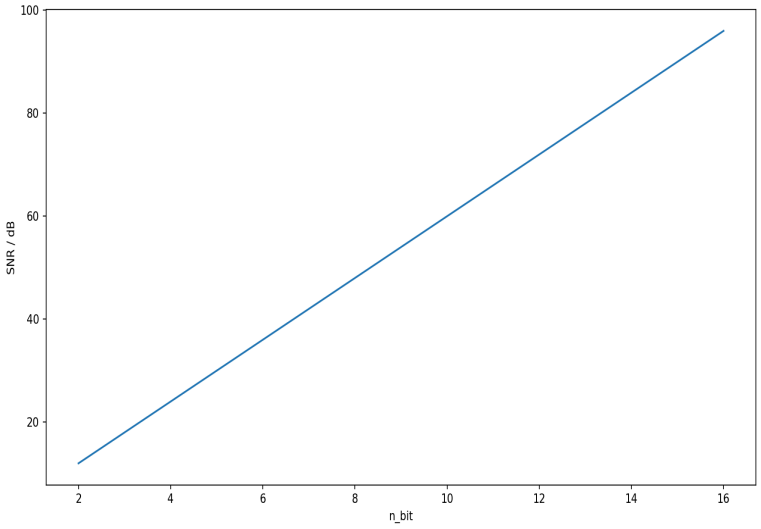
- **Tools and Libraries:** Python was used as the primary programming language to define ADC, BSC, and PCM classes. We used the numpy library to handle arrays, and matplotlib.pyplot for generating plots.
- **Key Steps:** We defined the ADC class with an n_{bit} attribute and the BSC class with an $error_probability$ attribute. Methods named snr() were implemented in both classes to calculate their respective signal-to-noise ratios. A unified PCM class was created to model the entire block diagram cascade. Furthermore, we utilized the Shannon sampling theorem to calculate the necessary ADC sampling frequency, acting as a multiple of a base frequency step, to successfully reconstruct an analog signal.
- **Justification:** The object-oriented approach is ideal because it creates modular software abstractions that directly map to the distinct hardware blocks of a real PCM system. Additionally, numpy was specifically required to handle large arrays of parameter configurations while explicitly managing data types to prevent calculation errors.

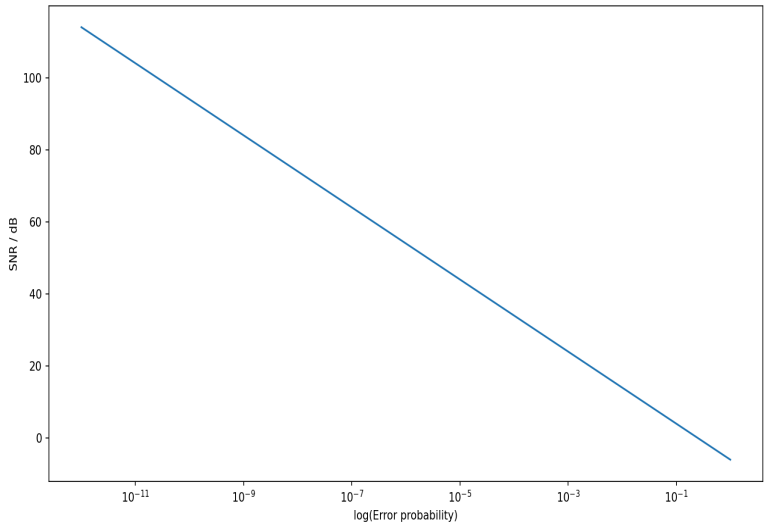
3. Results and Analysis

3.1. Individual Component Characterization (ADC and BSC)

The first phase of the simulation isolated the performance of the ADC and the BSC.

- **Quantization SNR vs. Number of Bits:** As expected, the relationship between the ADC's number of bits (n_{bit}) and the quantization SNR is linear. The generated plot demonstrates that increasing n_{bit} from 2 to 16 bits leads to a proportional increase in SNR, scaling from 12 dB to 96 dB (approximately). This confirms the standard rule of thumb of gaining approximately 6 dB of SNR per additional bit of quantization.
- **BSC SNR vs. Error Probability:** The simulation of the isolated channel noise demonstrates an inverse logarithmic relationship. At extremely low error probabilities (e.g., 10^{-11}), the channel supports a highly theoretical SNR of over 100 dB. As P_e approaches 10^{-1}, the SNR degrades rapidly, eventually crossing below 0 dB, indicating that the noise has completely overpowered the signal.

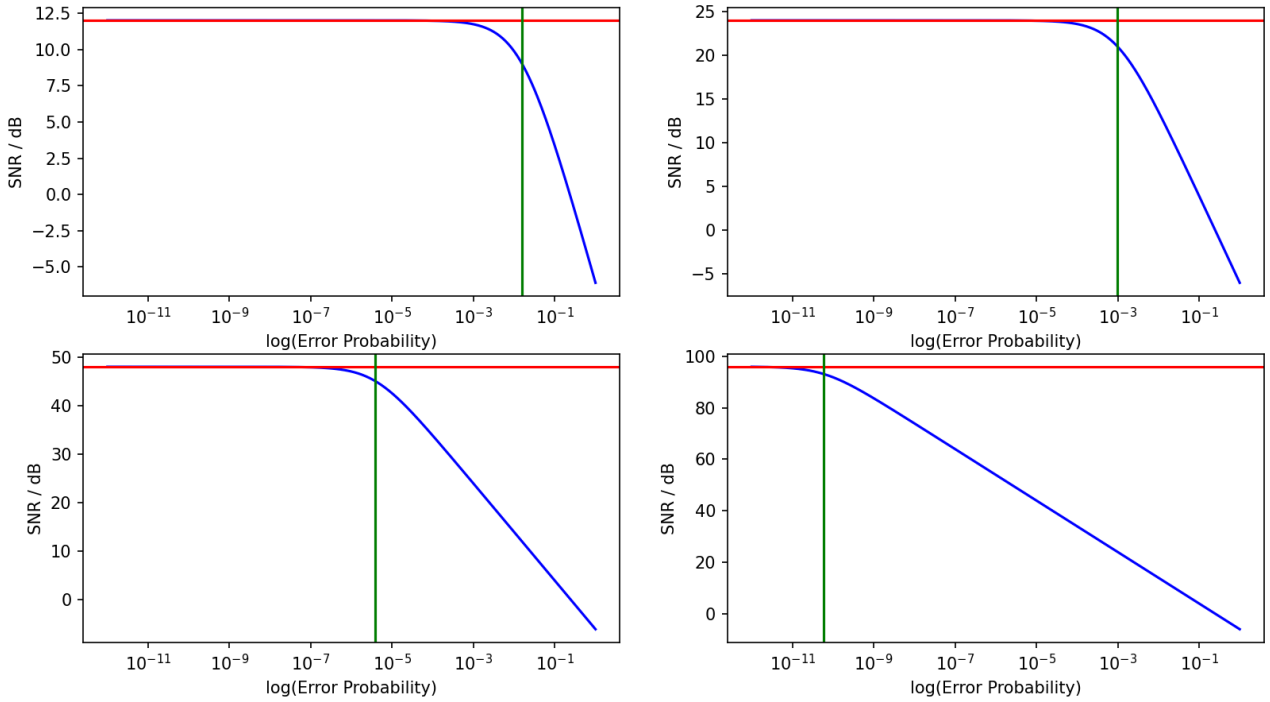




3.2. Overall PCM System Performance

To analyze the PCM diagram, the overall system SNR was calculated as using the following formula: $\frac{1}{SNR_{PCM}} = \frac{1}{SNR_{BSC}} + \frac{1}{SNR_{ADC}}$ for different n_{bit} configurations ($n_{bit} = 2, 4, 8, 16$). The resulting 2x2 subplot grid reveals distinct operational regions:

- **Quantization-Limited Region:** For low error probabilities (the left side of each subplot), the total SNR curve (blue line) remains perfectly flat and aligns with the quantization SNR (horizontal red line). In this state, the channel is relatively clean, and the signal quality is entirely bottlenecked by the n_{bit}.
- **Channel-Limited Region:** As the error probability increases, the blue curve breaks away from the red line and drops off sharply. In this state, the transmission errors introduced by the BSC (e_B) become the main source of noise, while ADC's precision is irrelevant.
- **Critical Threshold:** The vertical green line on each plot marks the critical error probability (P_{th}). This visualizes the transition point where the BSC noise is equal to the quantizing noise. As n_{bit} increases across the subplots, the critical P_{th} shifts to the left, demonstrating that high-resolution signals are far more sensitive.

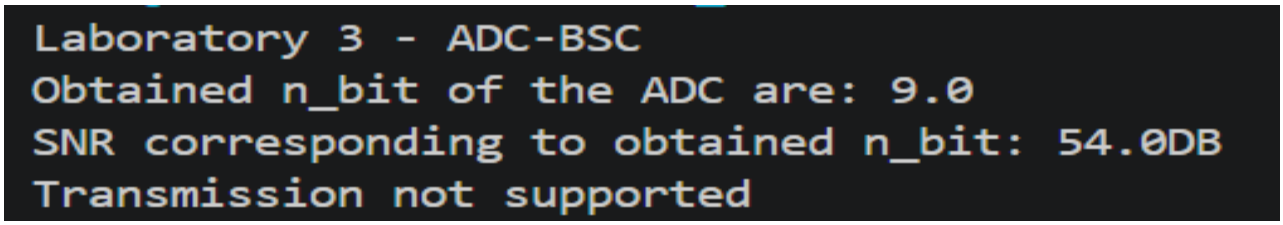


3.3. Design Application and System Constraints

The final phase of the experiment tested a specific application scenario (a 22 KHz bandwidth signal transmitted over a channel with $P_e = 3.8 \times 10^{-7}$).

Based on the simulation terminal output, the system evaluated the parameters with the following results:

- **Calculated ADC Bit Depth:** $n_{bit} = 9.0$
- **Effective Quantization SNR:** 54.0 dB
- **Transmission Status:** Not Supported



Analysis: While the initial goal in the lab instructions aimed for a SNR_{ADC} of at least 80 dB, using only $n_{bit} = 9dB$ provides us with SNR_{ADC} of 54dB. Due to this, last line of the code output is "Transmission not supported" correctly indicates that provided SNR_{ADC} is lower than the one required for the transmission. Yet, even if a sufficiently high n_{bit} was used, it would not make the transmission possible. This is due to the limit of the SNR_{BSC}, due to high error probability. In this case SNR_{BSC} is the main source of noise, and this validates that a PCM system is only as strong as its weakest system.

4. Challenges and Solutions

- **Numerical Overflow:** Handling array-based calculations for wide ranges of bit configurations posed a risk of number representation overflow. This was resolved by specifically setting the numpy array dtype parameter to int64, so each number uses 64bits, to ensure no overflow.
- **Data Visualization:** Plotting error probabilities across vastly different magnitudes (from 1×10^{-12} to 1) made it difficult to analyze behavior linearly. This challenge was overcome by configuring the x-axis to a logarithmic scale using the plt.xscale("log") function.

5. Connection Between Theory and Practice

- **Reinforcing Concepts:** The lab confirms the theoretical formula that total system SNR can be modeled mathematically as $\frac{1}{SNR_{PCM}} = \frac{1}{SNR_{BSC}} + \frac{1}{SNR_{ADC}}$.
- **Insights Gained:** By plotting vertical horizontal lines for theoretical thresholds, the simulation visually demonstrates the critical error probability (P_{th}). This visualization allows us to see exactly where the transmission channel's noise (e_B) overtakes the quantization noise (e_Q) as the limiting factor in signal transmission.

6. Conclusions

- **Key Findings:** The lab successfully constructed a working software representation of a PCM system that includes an ADC, BSC, and DAC. The simulation confirms that the required sampling frequency and the ADC bit depth must be configured to meet minimum output SNR requirements. It also highlights if a given channel's error probability can successfully support the transmission of a digitized signal without a great quality loss.
- **Future Work:** Possible improvements could involve removing the assumption of perfect analog signal reconstruction at the DAC. This would make the PCM representation much more realistic and yield more realistic results in the end.